

Joseph Burton

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EXPERIENCE

Bechtel Corporation | Reston, VA

December 2025 – Present

Pipe and Valve Engineer

- Natrium Demonstration Project
 - Piping component design consisting of wall thickness calculations and piping component specification
 - Procurement of pipe, piping components, and valves including communication with vendors, technical specifications, and technical bid evaluation

UVA Fluids Research and Innovation Laboratory | Charlottesville, VA

May 2024 – December 2025

Researcher

- Compressed Air Energy Storage (CAES)
 - Experimental setup, operation, and data processing for a lab-scale piston
 - Developed a theory to predict round-trip efficiency of a piston based on its operating parameters
- Floating Offshore Wind Turbine (FOWT)
 - Collected and processed test data for scaled-down models of FOWT designs

Laser & Plasma Technologies | Charlottesville, VA

January 2024 - March 2024

Intern

- Research and development
 - Contributed to a NASA project involving the development of an optic fiber-based hybrid spectroscope

M.C. Dean | Tysons, VA

June 2023 - August 2023

Engineering Intern

- Worked on a modular design construction project
 - Extensive work in SolidWorks with experience in sketching, weldments, assemblies, and drawings

UVA Undergraduate Physics Lab | Charlottesville, VA

October 2022 - May 2023

Undergraduate Research Assistant

- Research on magnetic domains in thin films
 - Created scripts in python that used microscopic images of magnetic domains to quantify properties of these domains

EDUCATION

University of Virginia | Charlottesville, VA

Graduate: Mechanical and Aerospace Engineering (M.E.)

Graduated May 2025 | GPA: 3.85/4.0

Undergraduate: Mechanical Engineering (B.S.), Physics (B.A.)

Graduated May 2024 | GPA: 3.70/4.0

Fairfax High School | Fairfax, VA

Advanced High School Diploma

Graduated June 2020 | GPA: 4.51/4.00

SKILLS SUMMARY

- Embedded microcontroller programming with Arduino and Propeller
- CAD modeling using SolidWorks, Autodesk Inventor, and FreeCAD; 3D printing and prototyping
- Experience in engineering thinking and calculations, e.g. modeling heat transfer, fluid mechanics, stresses, etc.
- Script writing in MATLAB, Python, C, and R
- Technical writing